

Name: _____ score: _____ / 10

There are 10 points possible on this quiz. No aids (book, notes, etc.) are permitted. You may use a non-programmable calculator. **Show all work and supporting calculations for full credit. Explain how you get your answers.**

1. Alden and Bobby buy a large pizza together for \$32. It is split equally into one-third veggie, one-third anchovy, and one-third pineapple.

- (a) (2 points) Alden doesn't like anchovies, so he values the anchovy part at \$0, and he values pineapple and veggie equally. Bobby values the veggie **twice** as much as the anchovy and he values the pineapple **five times** as much as the anchovy. Complete the following table to indicate the value each party would assign to each part of the pizza.

party	veggie	anchovy	pineapple	total value
Alden	16	0	16	32 ✓
Bobby	8	4	20	32 ✓

try:
$$\begin{array}{ccc} V & A & P \\ \hline 2 & 1 & 5 \end{array} \quad \begin{array}{c} \text{total} \\ 8 \end{array} \rightarrow 32$$

$32/8 = 4$, multiply by 4

- (b) (3 points) Suppose they split the pizza using the **Divider-Chooser** method with Alden as the divider. He divides the pizza into Share 1, which consists of the anchovy and veggie portions of the pizza and Share 2, which consists of the pineapple portion of the pizza. Complete the Divider-Chooser method, including the final outcome for each party. Be sure to fully justify your answer and verify that each party ends up with a fair share.

Bobby values

Share 1: $A + V = 4 + 8 = 12$

Share 2: $P = 20$

Fair Share = $\frac{32}{2}$
= 16

Bobby values share 2 as a fair share, worth \$20

Alden gets Share 1, worth \$16

both are ≥ 16 so fair shares to each

Alden's final outcome 81 Bobby's final outcome 52

2. Linda, Ned, and Sophie will split a bag of jelly beans worth \$30 using the **Lone Divider** method. One of these people is selected to divide the jelly beans into three shares. The values are shown below.

Person	Share 1	Share 2	Share 3
Linda	\$8	\$15	\$7
Ned	\$10	\$10	\$10
Sophie	\$5	\$20	\$5

- (a) (1 point) What is the value of a fair share to each person?

$$\frac{\$30}{3 \text{ people}} = \$10 = \text{fair share}$$

- (b) (1 point) Who was the divider? How can you tell?

Ned, he values each share equally

- (c) (1 points) Circle the bids (dollar values) in the table that are fair shares to each person.

done

- (d) (2 points) State the final allocation of the shares or describe what happens next in the Lone Divider process.

There's no way to allocate shares so that each person is happy because Linda & Sophie only value Share 2 as a fair share.

So we allocate an uncontested share (say share 1) to Ned and then Linda & Sophie do Divider-Chooser on the remainder.